Intern Logbook

October 6 University Faculty of Applied Health

Science Technology

Biotechnology Specialization



2026-2027

Intern Logbook
October 6 University
Faculty of Applied Health Science Technology
Biotechnology Specialization

Photo

Student Name:.....

Student ID:.....

Academic Year:.....

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Introduction

It is a structured book where certain educational activities and patient-related information are recorded, usually by hand. Logbooks are widely used in practical and hands-on training programs in pharmaceutical research and development, natural products, cell culture, and computational drug discovery, either in paper or electronic format.

Logbooks provide a clear setting of learning objectives and give trainees and clinical teachers a quick overview of the training requirements, as well as an idea of their learning progress. Logbooks are especially useful if different sites are involved in the training to set a standard of training.

Logbooks assist supervisors and trainees to see at one glance which learning objectives have not yet been accomplished and to set a learning plan. The analysis of logbooks can reveal weak points in training and evaluate whether trainees have fulfilled the minimum training requirements.

Logbooks facilitate communication between the trainee and clinical teacher. It also helps to structure and standardize learning in clinical settings. In contrast to portfolios, which focus on students' documentation and self-reflection of their learning activities, logbooks set clear learning objectives to help structure the learning process in clinical settings and to ease communication between trainee and clinical teacher.

Training Objectives

1- Opening communication channels between college students and the labor market on the one hand and between faculty members, college administration, and the labor market on the other hand, and what this may achieve in terms of developing study plans in each specialty.

2- Providing the student with the opportunity to gain practical experience before graduation.

3- Accustom the student to taking responsibility, listening to the opinions of others, and using other teamwork skills.

4- Teaching students time management skills.

5- Deepening students' understanding of the theoretical sciences they have received in their academic studies.

6- Introduce the student to the fields of work he is expected to work in after graduation, and experience the actual working conditions.

7- Introducing the training authorities to the level of students in the college facilitates the process of attracting and employing them.

8- Enhance skills in the identification of anatomical structures and normal findings.

9- Develop an understanding of a laboratory's legal and ethical considerations.

10- Learn to work as part of a multidisciplinary team, collaborating with other healthcare professionals to provide optimal patient care.

11- Enhance diagnostic interpretation skills by analyzing a diverse range of normal and abnormal conditions.

Pharmaceutical Biotechnology & Drug Discovery for 4th-year trainees

Biotechnology Competencies/Skill/Procedure

A laboratory technologist specialist must possess a range of varied and complex clinical competencies. Below is a complete list of the skills necessary for trainees and trainers.

The level of competence to be achieved each year is specified according to the key as follows:

1. Observer status
2. Assistant Status
3. Performed under supervision.
4. Performed under indirect supervision.
5. Performed independently

Part 1 : Phytochemistry

Trainee's Tasks:

By the end of working in this section, the student should be able to:

1. Identify and appropriately handle plant materials used for phytochemical investigation.
2. Apply basic principles of plant material collection, preparation, drying, and storage.
3. Perform suitable extraction procedures for obtaining bioactive compounds from plant materials.
4. Understand the principles of solvent selection and extraction efficiency.
5. Differentiate between primary and secondary metabolites and their biological relevance.
6. Perform preliminary phytochemical screening for major classes of secondary metabolites.
7. Interpret preliminary screening results to identify potentially bioactive extracts.
8. Select suitable extracts for further phytochemical separation and biological evaluation.
9. Understand the principles of separation and purification of phytochemicals.
10. Apply thin-layer chromatography (TLC) for monitoring and separating phytochemical components.
11. Understand the basic principles and applications of chromatographic techniques in phytochemical analysis.
12. Understand the basic principles of HPLC and its application in phytochemical analysis.
13. Perform basic preparation and handling of samples for chromatographic analysis.
14. Interpret chromatographic profiles and compare phytochemical compositions between samples.
15. Understand the basic principles of spectroscopic characterization of phytochemicals.
16. Relate chromatographic and spectroscopic data to the identification and characterization of bioactive compounds.
17. Evaluate analytical results to support the selection and characterization of phytochemical candidates.

Lab 1:Phytochemitry

Duration: From ……..…… to……..

Procedure	Date	Level of Training	
		Observe	Perform
Introduction to Natural Products and Phytochemical Research			
Plant Material Collection, Preparation, Drying, and Storage			
Selection of Solvents and Extraction Methods			
Extraction of Bioactive Compounds from Plant Materials			
Preliminary Phytochemical Screening			

Supervisor
Dr Ayman Saleh

Training Entity
Creative Egyptian Biotechnologists - CEB

Lab 2:Phytochemitry

Duration: From ……..…… to……..

Procedure	Date	Level of Training	
		Observe	Perform
Principles of Phytochemical Separation and Purification			
Thin-Layer Chromatography (TLC) and Chromatographic Separation			
HPLC Principles and Applications in Phytochemical Analysis			
Spectroscopic Characterization of Phytochemicals			
Interpretation of Chromatographic and Spectroscopic Results			

Supervisor
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Part 2 :Cell Culture

Trainee's Tasks:

By the end of working in this section, the student should be able to:

1. Understand the basic principles and importance of animal cell culture in biomedical and biotechnological research.
2. Identify different types of cell culture systems (primary cell culture and secondary cell lines).
3. Perform aseptic techniques required for handling cells and maintaining a sterile environment.
4. Prepare and use appropriate culture media, supplements, and buffers for cell growth.
5. Subculture (passage) adherent cells using proper enzymatic methods.
6. Observe cell morphology, confluency, and contamination under an inverted microscope.
7. Safely dispose of biological waste and follow biosafety level requirements for cell culture laboratories.

Lab 1: Cell Culture

Duration: From ……..…… to……..

Procedure	Date	Level of Training	
		Observe	Perform
1. Introduction to mammalian cell culture			
2. Cell Passaging & Proliferation			
3. Assessing confluency and cell morphology.			
4. Subculture techniques			
5. Discard the old medium, wash the cells with PBS, add trypsin (Trypsinization)			

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Lab 2: Cell Culture

Duration: From ……..…… to……..

Procedure	Date	Level of Training	
		Observe	Perform
1. Cell counting using a hemocytometer			
2. Mix cell suspension with Trypan Blue and load10 µL of the mixture onto the hemocytometer			
3. differentiate viable from non-viable cells			
4. count the cells and calculate the cell concentration			
5. Use the cell counting results to Cell Seeding Calculation			
6. calculate the required number of cells and the volume of cell suspension needed per well			
7. Seed the cells at the required density and passage number			
8. incubate under the experimental conditions to monitor cell attachment			

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Lab 3:Cell Culture

Duration: From ……..…… to……..

Procedure	Date	Level of Training	
		Observe	Perform
1. Experimental design for evaluating extracts and compounds			
2. Prepare the required drug concentrations using serial dilution			
3. MTT cell viability and cytotoxicity assay			
4. Add 20 µL of the prepared drug dilutions to the designated wells and incubate the test plate at 37°C and 5% CO_2			
5. Add MTT reagent to each well s, add DMSO to dissolve the formazan crystals			
6. Measure absorbance at 570 nm using a microplate reader.			
7. IC_{50} calculation and interpretation			
8. Data Analysis & interpret the results by comparing treated cells with control cells.			

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Part 3 : Drug Discovery

Trainee's Tasks:

By the end of working in this section, the student should be able to:

1. Understand the drug discovery and development pipeline.
2. Differentiate between hit and lead compounds.
3. Understand the principles of target identification and validation.
4. Connect observed biological activity with potential molecular targets.
5. Understand basic protein structures and ligand preparation.
6. Describe protein–ligand interactions.
7. Understand the principles of virtual screening.
8. Understand the molecular docking workflow.
9. Interpret docking scores and analyze molecular interactions.
10. Evaluate and prioritize potential drug candidates.
11. Understand the basic concepts of drug-likeness and ADME assessment.

Lec 1: Drug Discovery

Duration: From ……..…… to……..

Procedure	Date	Level of Training	
		Observe	Perform
Overview of the drug discovery and development pipeline			
Target identification and target validation			
Drug targets and their relevance to disease mechanisms			
Concepts of hits, hit identification, and lead compounds			
Lead optimization and candidate selection			
Connecting observed biological activity with potential molecular targets			
Introduction to target-based and phenotypic drug discovery approaches			
Integration of biological, biochemical, and pharmacological evidence in drug discovery			
Retrieval and examination of experimentally determined protein structures using the Protein Data Bank (PDB).			
Retrieval of known bioactive compounds and ligand structures using PubChem and ChEMBL.			

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Lec 2: Drug Discovery

Duration: From ……..…… to……..

Procedure	Date	Level of Training	
		Observe	Perform
Molecular docking workflow, including receptor and ligand preparation, docking setup, and pose generation			
Preparation and inspection of protein structures for computational analysis			
Ligand structure preparation and optimization for computational drug discovery.			
Identification and analysis of protein binding sites and potential ligand-binding regions			
Interpretation of docking scores and analysis of predicted binding poses and molecular interactions			

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Lec 3: Drug Discovery

Duration: From ……..…… to……..

Procedure	Date	Level of Training	
		Observe	Perform
. Introduction to drug-likeness and ADME assessment and their role in evaluating potential drug candidates.			
Basic ADME concepts and their relevance to drug candidate selection			
Project (isease → Target identification → UniProt/NCBI → PDB → Ligand identification → PubChem/ChEMBL → Protein & ligand preparation → Docking → Interaction analysis → Drug-likeness/ADME)			

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